

OAKRIDGE ACADEMY OF SCIENCE & TECHNOLOGY

Department of Physics & Chemical Sciences

Problem Set #4: Integrated STEM

Course: AP Physics & Chemistry | Instructor: Dr. E. Vance

TOTAL POINTS: 100

Score: ____ / 100

Due: Oct 28, 2026

✓ **Calculators:** Scientific/graphing allowed (TI-84 style).

✓ **Show Work:** Full credit requires setup, values, and units.

✓ **Integrity:** Concepts may be discussed; work must be unique.

✓ **Submission:** Place in Bin B by 3:15 PM on due date.

1 Section I: Multiple Choice & Conceptual Analysis (20 Points)

Select the best response for each item. Encircle your choice and write a brief 1-sentence justification.

- Q1.** A monochrome beam of green laser light ($\lambda = 532 \text{ nm}$) transitions from air ($n = 1.00$) into crown glass ($n = 1.52$). Which statement correctly describes the change in the wave's properties?

- (A) The frequency decreases, while the wavelength remains constant.
 (B) The frequency remains constant, while the wavelength decreases to approximately 350 nm.
 (C) The speed increases to $4.56 \times 10^8 \text{ m/s}$, and frequency increases.
 (D) Both speed and frequency decrease by a factor of 1.52.

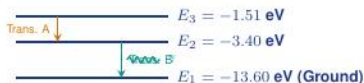
Justification: _____

- Q2.** A 2.50 kg aluminum block (90.0°C) is placed into 1.00 kg of water (20.0°C) in an insulated calorimeter. What is the equilibrium temperature T_f ? ($c_{\text{Al}} = 900 \text{ J/(kg} \cdot ^\circ\text{C)}$, $c_w = 4184 \text{ J/(kg} \cdot ^\circ\text{C)}$).

- (A) 35.0°C (B) 44.5°C (C) 55.0°C (D) 68.2°C

Justification: _____

- Q3.** Refer to the energy level schematic for synthetic atom **Vertex-9**. A photon of frequency $f = 6.18 \times 10^{14} \text{ Hz}$ is emitted during electronic relaxation.



Which quantum transition generated this emitted photon?

- (A) $E_3 \rightarrow E_2$ (B) $E_2 \rightarrow E_1$ (C) $E_3 \rightarrow E_1$ (D) $E_1 \rightarrow \infty$

Justification: _____

- Q4.** The synthesis of ammonia ($\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$) has $E_a = 145 \text{ kJ/mol}$. When a catalyst (**Nexa Corp**) is added:

- (A) ΔH° becomes more negative.
 (B) Both forward and reverse E_a are lowered by equal energy amounts.
 (C) The equilibrium yield of NH_3 increases at constant temperature.
 (D) Forward E_a decreases while reverse E_a increases.

Justification: _____